Towards atoms in a fibre

Modulating, multiplexing and memorising photons for quantum networks and computing

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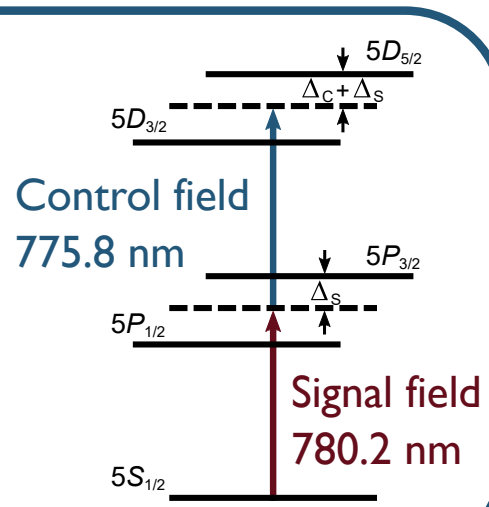
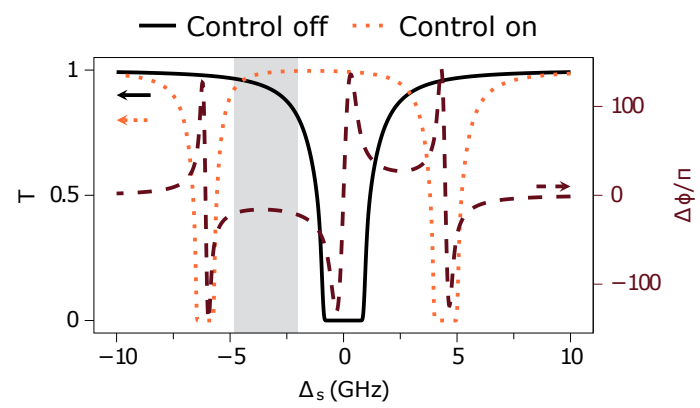
We demonstrate all optical phase modulation mediated by a two-photon transition in warm atomic vapour, alongside a fibre-integrated vapour cell with low-loss interconnects.

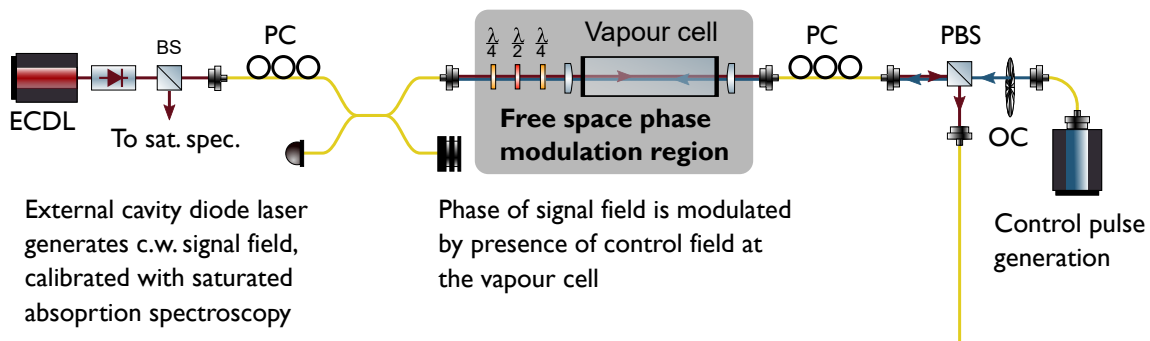
1. Phase modulation

- A weak signal field counter-propagates with a strong control pulse through a rubidium vapour cell.
- The presence of the control pulse induces a change in susceptibility, resulting in a change to the phase of the signal.

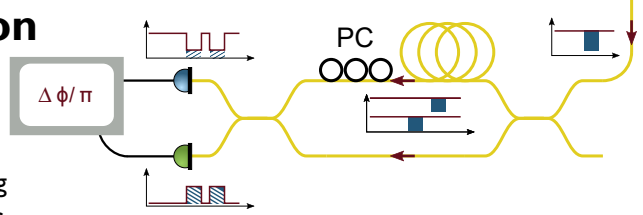




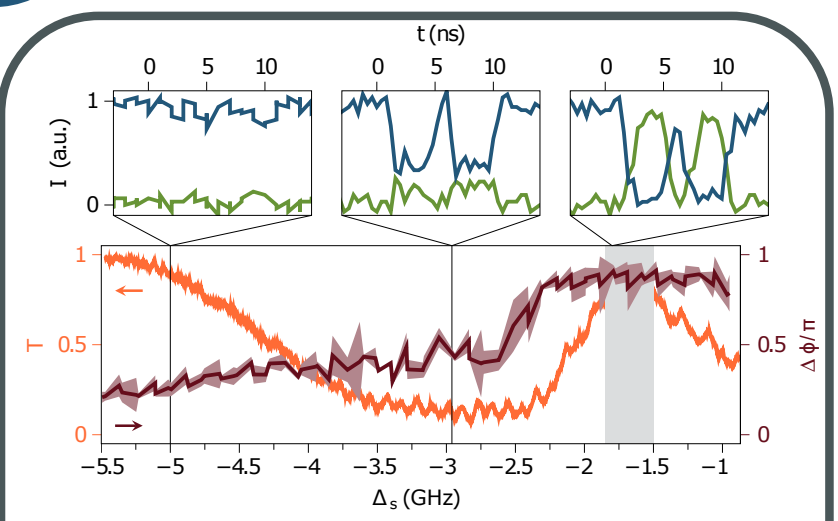
We model phase shift by solving the Maxwell-Bloch equations of motion.



Signal detection



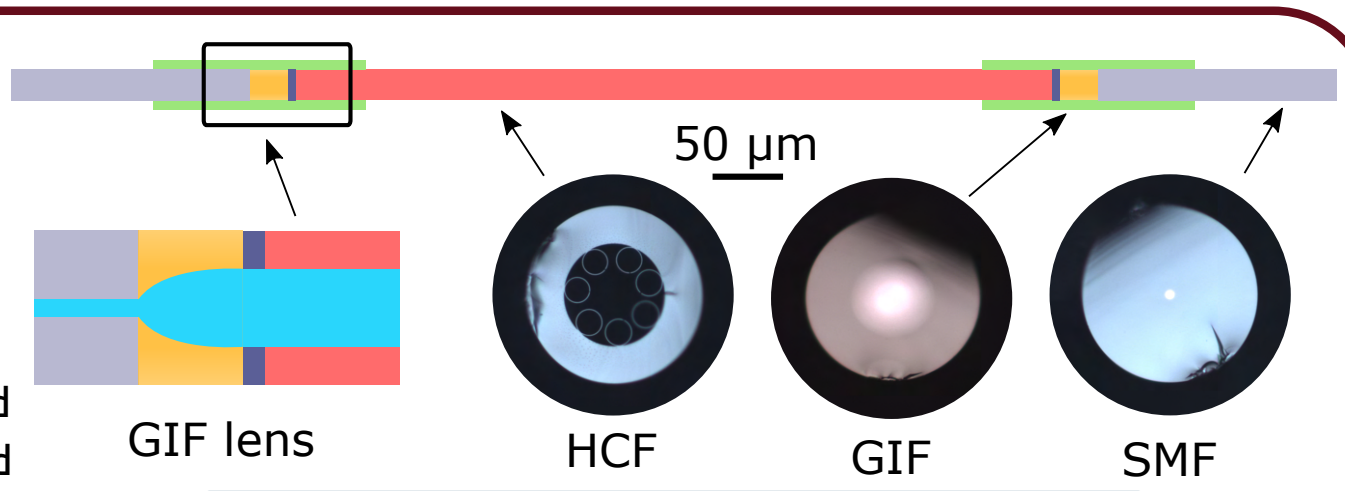
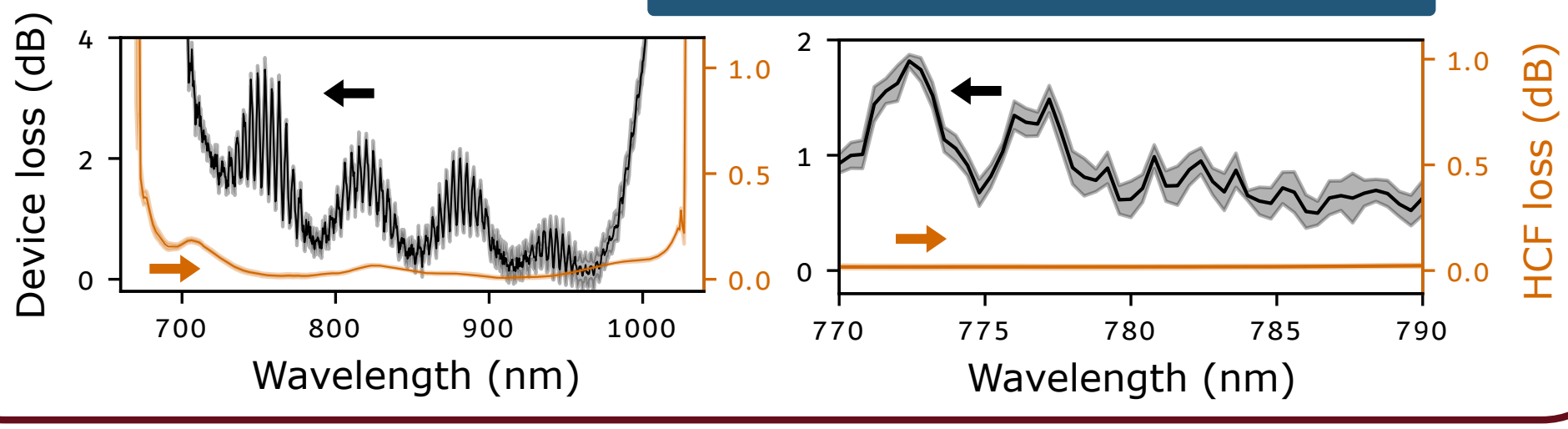
Phase shift is detected using time-binned interferometer



High phase shift with low loss: $\Delta\phi/\pi = 0.90\pm0.05$; $T = 74\pm2$ %

2. Fibre integration

- We have fabricated a cell for realising phase modulation and memories within fibre.
- Low-loss interconnects to HCF are achieved by lensing in graded index fibre into custom-designed hollow fibre.



Low insertion loss: 0.6(2) dB at 780nm